

YOUR OFE 2025 REPORT

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2025 Cropping Season

Field M5F2S2 | NRate 163 lb N/ac | Field monitoring report — single N rate, no treatment split

Welcome to your 2025 farm report. This report summarizes field-season measurements for the **M5F2S2** field. Because this field ran under a single nitrogen rate of 163 lb N/ac with no treatment or control strips, we're not comparing two systems here, instead, we're giving you a field-level picture of how the corn crop performed, what the cover crop residue looked like going into the season, and what the soil health assessment tells us about the foundation you're working with. Think of this as your baseline.



Figure 1. M5F2S2 field.

Take-home message: The corn crop across this field showed consistent nitrogen status mid-season, with tissue nitrogen averaging 2.76% across the sampled points at the 163 lb N/ac input rate. The cover crop residue had a C:N ratio of 22.6, low enough to release nitrogen relatively quickly early in the season, which is a good sign for future nitrogen credit potential. The main concern from the soil health assessment is surface compaction, scored 10/100, which is worth addressing before it limits rooting and water movement long-term. Chemistry is excellent across the board. A clean growing season with no drought stress during the critical window rounded out a solid year.

Results Summary

Metric	Value	SD
 Maize Biomass - Mean %N	2.764%	± 0.179
 Maize Biomass - C:N Ratio	16.28	± 1.14
 Cover Crop - Mean %N	2.024%	± 0.515
 Cover Crop - C:N Ratio	22.63	± 5.06
 Soil Health Score	62 / 100	—

This field did not have a treatment/control split. All statistics are descriptive field-level observations. No statistical comparisons between groups are reported.

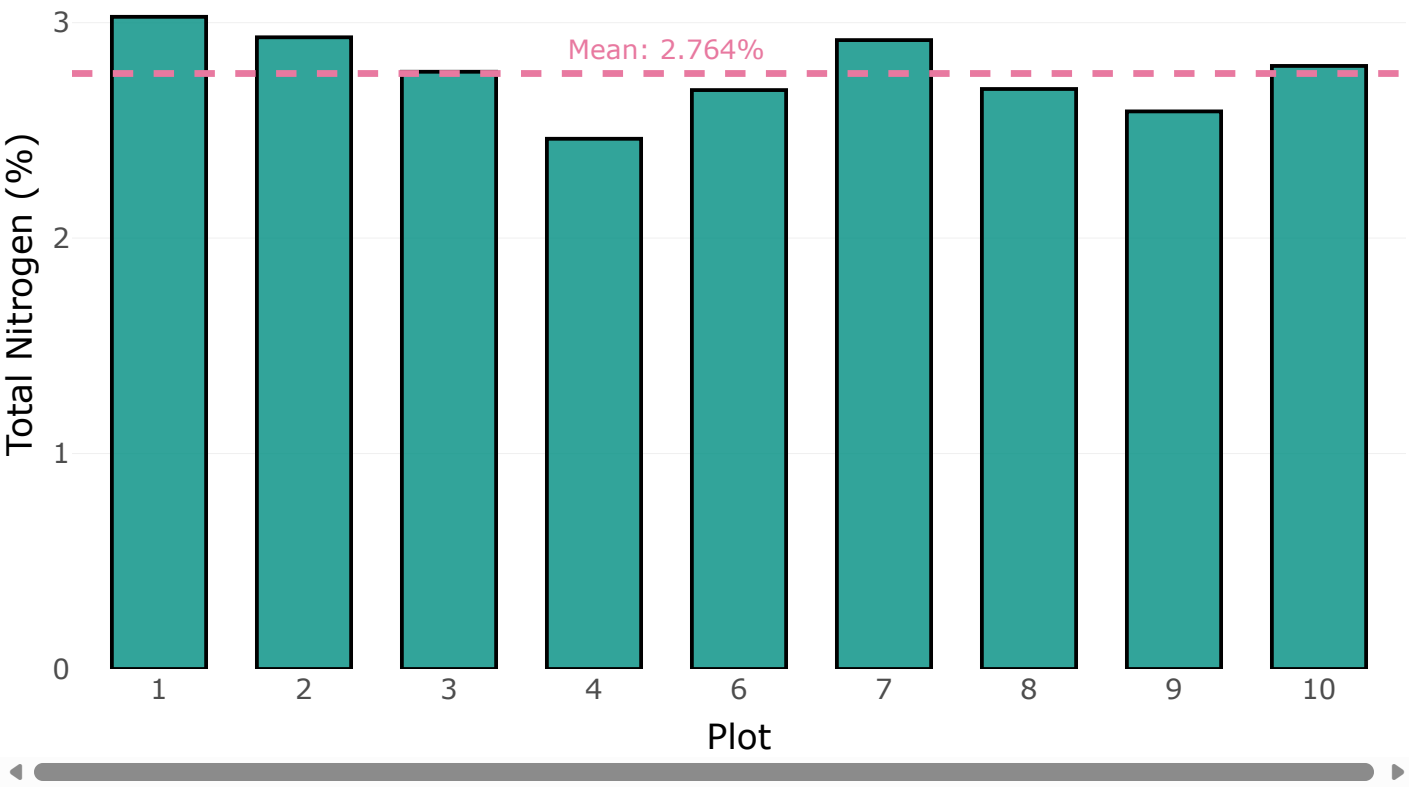
C:N in Corn Biomass

Corn biomass was collected from **9 points** across the North field and sent to the laboratory for C:N analysis. This tells us how much nitrogen the plant had absorbed and incorporated into its tissue mid-season, a direct indicator of nitrogen sufficiency at the 163 lb N/ac input rate. Because there is no treatment split, we describe the **field-level distribution and spatial variability** across points.

Total Nitrogen in Biomass (%N)

Bar Chart (%N by plot)

Total Nitrogen in Corn Biomass by Plot (%N)



C:N Ratio — Biomass

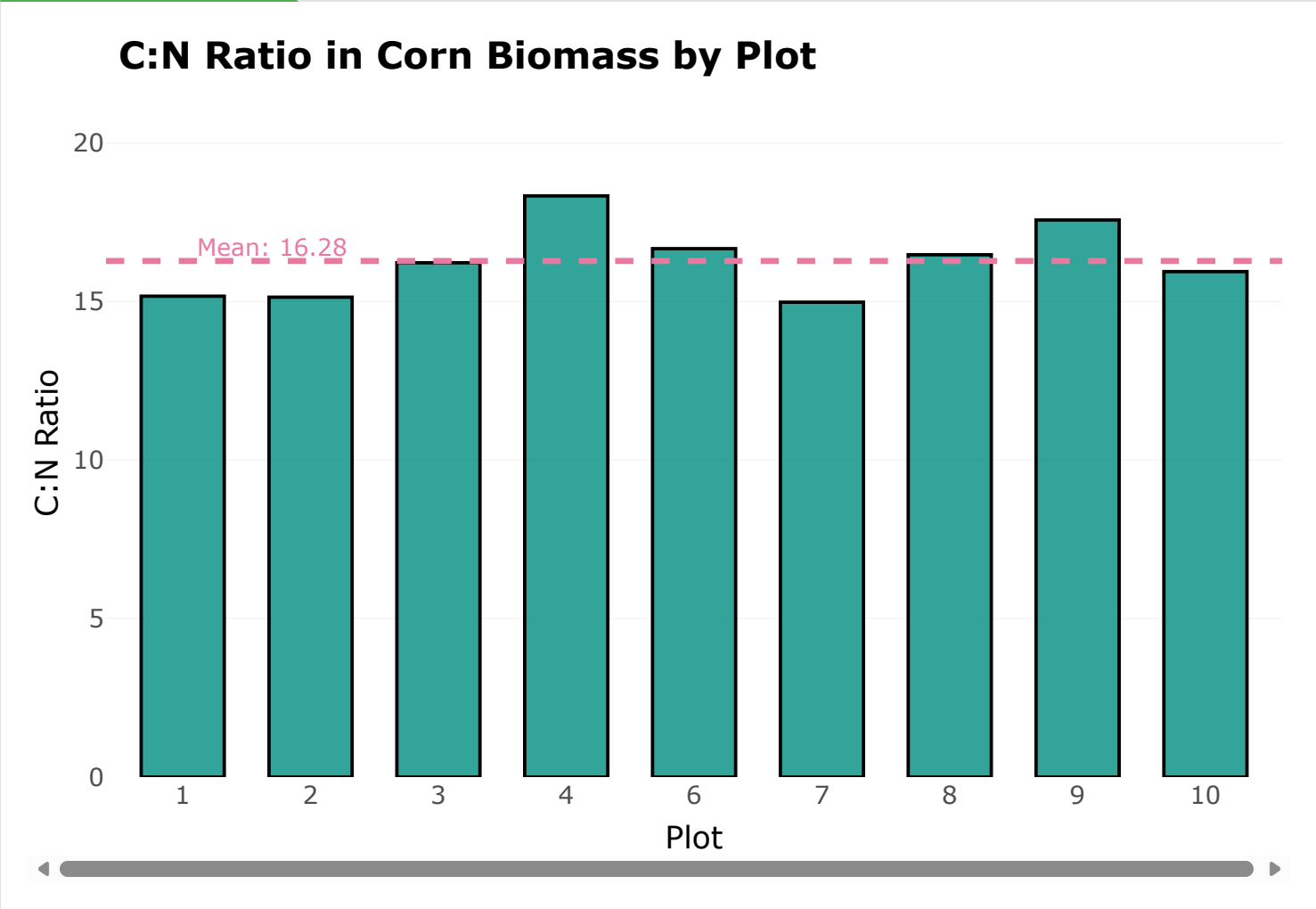
MEAN C:N — BIOMASS

16.28

SD: 1.14 | SE: 0.38

Bar Chart (C:N by plot)

Scatter %C vs %N



Cover Crop C:N

5 cover crop samples were collected from field M5F2S2 and analyzed for C:N content. The C:N ratio of cover crop residue tells us how quickly it will break down and release its nitrogen to the following corn crop. A C:N below 25 means decomposition happens relatively fast, the nitrogen becomes available early in the season, right when the young corn plant is establishing its root system and building its nitrogen demand. The bar charts below show the %N and C:N ratio for each sample location.

MEAN %N — COVER CROP

2.024%

SD: 0.515 | n = 5 samples | Range: 1.46 – 2.87%

MEAN C:N — COVER CROP

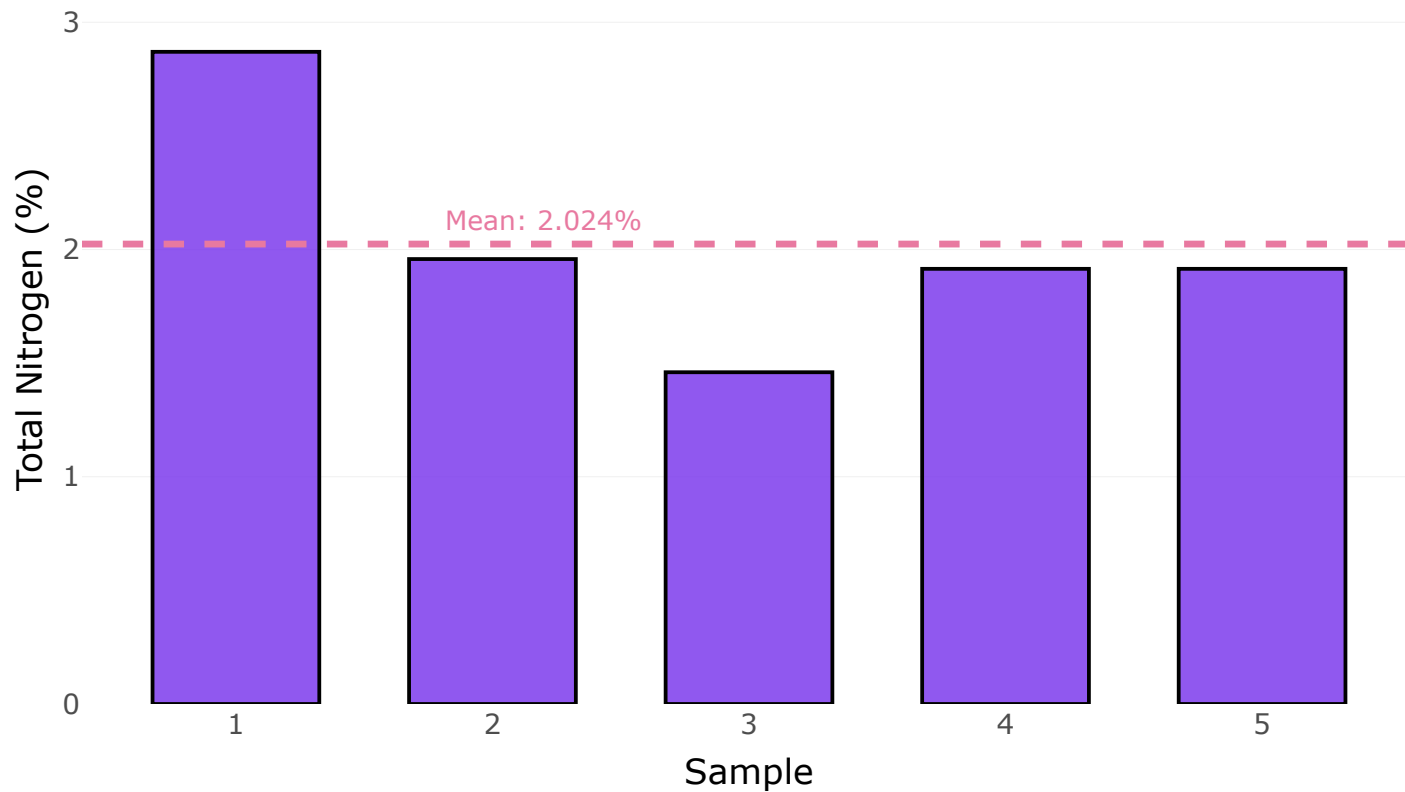
22.63

SD: 5.06 | low C:N — fast decomposition, good early-season N supply

Total Nitrogen in Cover Crop (%N)

C:N Ratio — Cover Crop

Total Nitrogen in Cover Crop (%N)



Soil Health Assessment

Soil health was assessed using the **Cornell Soil Health Test**. The overall quality score was **62/100 (High)**. Chemistry is in excellent shape across the board: pH at 6.5, phosphorus, potassium, and all additional nutrients all came back at 100. The area that needs attention is physical structure. Surface hardness scored 10/100, the only red indicator in the entire report, meaning the top layer of your soil is compacted enough to restrict root growth and slow water movement through the profile. Biological indicators are in the low-to-medium range: soil respiration, predicted soil protein, and aggregate stability all have room to improve. These are the indicators that respond best to reduced tillage and continuous cover cropping over time.

OVERALL QUALITY SCORE

62 / 100

Rating: High | Cornell Soil Health Lab

SURFACE HARDNESS

10 / 100

Very Low - constraining rooting & water transmission

Comprehensive Assessment of Soil Health

From the Cornell Soil Health Laboratory, Department of Soil and Crop Sciences
School of Integrative Plant Science, Cornell University, Ithaca, NY 14853
<https://soilhealthlab.cals.cornell.edu>



Grower: Ron Grushow

Agricultural Service Provider:

Alex Yore
ay327@cornell.edu

Sample ID: QCC91
Field ID: OFE25 N AFTER CC GRUSHOW N
Date Sampled: 07/07/2025
Coordinates: Latitude: 42.902008
Longitude: -77.595683

Measured Soil Textural Class: **loam**

Sand: **38%** - Silt: **45%** - Clay: **18%**

Group	Indicator	Value	Rating	Constraints
physical	Predicted Available Water Capacity	0.21	79	
physical	Surface Hardness	273	10	Rooting, Water Transmission
physical	Subsurface Hardness	327	40	
physical	Aggregate Stability	23.3	33	
biological	Organic Matter Soil Organic Carbon: 1.71 / Total Carbon: 1.75 / Total Nitrogen: 0.17	2.9	47	
biological	Predicted Soil Protein	5.62	39	
biological	Soil Respiration	0.5	32	
biological	Active Carbon	591	69	
chemical	Soil pH	6.5	100	
chemical	Extractable Phosphorus	6.5	100	
chemical	Extractable Potassium	119.8	100	
chemical	Additional Nutrients Ca: 1427.1 / Mg: 199.3 / S: 5.7 Al: 11 / B: 0.16 / Cu: 0.07 Fe: 1.4 / Mn: 2.1 / Zn: 0.8		100	

Overall Quality Score: **62** / High

Figure 2. Cornell Assessment of Soil Healthreport - Grushow North.

[Open Full Soil Health Report \(PDF\)](#)

Weather & Drought Conditions

Cumulative precipitation at your field compared to the all-site average, alongside the weekly drought index from June through November 2025. A large single-day rain event hit in mid-June, around 65 mm, which pushed your cumulative total well above the all-site average heading into July. From there, rainfall stayed consistent and moderate through the summer, tracking closely with the network average. The drought index stayed at None through most of September, with no stress during vegetative growth or grain fill. It moved to Low in late September and to Mild in October, and your field finished the season slightly below the all-site average (about 425 mm vs 470 mm). One thing worth watching: given the low aggregate stability score (33/100), that heavy June rain event may have contributed to some surface crusting or runoff, something to monitor in future seasons.

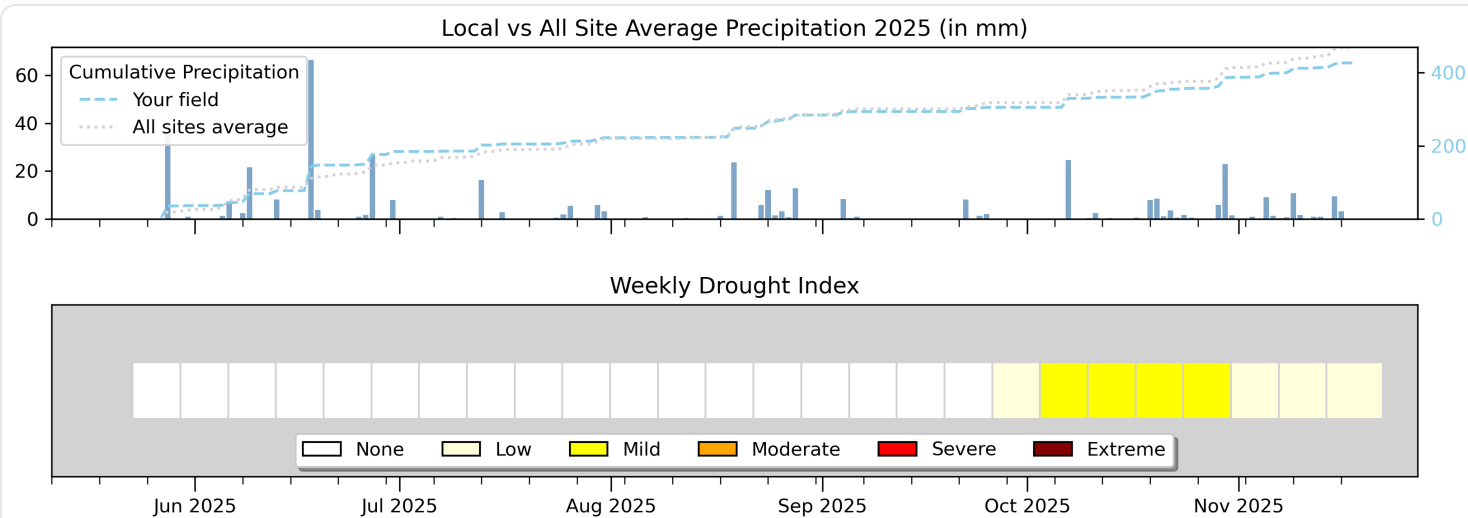


Figure 3. Local vs all-site average precipitation (top) and weekly drought index (bottom), June – November 2025.

Conclusions

MAIZE BIOMASS NITROGEN

Corn plants across the field averaged **2.76% N** in mid-season biomass, with a C:N ratio of **16.3**. Plot-to-plot variability was modest, tissue nitrogen was consistent across all 9 sampled locations at the 163 lb N/ac input rate. Without a comparison strip or a reference value for this growth stage, this number works best as a baseline for future seasons.

COVER CROP N CONTRIBUTION

The cover crop came in at a mean C:N of **22.6** and tissue N of **2.02%**, low enough to decompose relatively quickly and supply nitrogen.

SURFACE COMPACTION — PRIORITY CONSTRAINT

Surface hardness scored **10/100**. This level of compaction is restricting root growth and slowing water movement through the surface layer. Soil respiration (32), predicted soil protein (39), and aggregate stability (33) all came back in the Low range. The microbial community is working below its potential. Reducing tillage intensity, keeping the soil covered as long as possible, and adding fresh organic matter are the primary ways to move these numbers up over time.